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AND DISASTER RELIEF ASSISTANCE

Office of the United Nations Disaster Relief Co-ordinatorReport of the Secretary-General

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I. INTRODUCTION

1. The present report covers the work and activities of the Office of the United Nations Disaster Relief Co-ordinator (UNDRO) for the two years 1986-1987. It is presented to the General Assembly in response to General Assembly resolutions 2816 (XXVI) of 14 December 1971 and 39/217 of 18 December 1984, Assembly decision 40/436 of 17 December 1985 and Assembly resolution 41/201 of 8 December 1986.

2. It will be recalled that in General Assembly resolution 41/201, which deals with the work and activities of UNDRO, the Assembly requested the Secretary-General to submit to the General Assembly at its forty-second session a comprehensive review and assessment of the existing mechanisms and arrangements within the system for disaster and emergency assistance and co-ordination. The review was presented by the Secretary-General in his report (A/42/657), which contained recommendations both with regard to UNDRO operations and to arrangements at United Nations Headquarters for monitoring all types of emergencies and disasters. The General Assembly subsequently endorsed these recommendations in decision 42/433 of 11 December 1987 and will further examine their implementation during the second regular session of 1988 of the Economic and Social Council. At the forty-third session of the General Assembly, Member States will have before them a separate report on the issues covered in Assembly decision 42/433, which are of a system-wide nature and go beyond the work reported hereunder.

3. During the biennium, the activities of UNDRO were carried out in accordance with the medium-term plan for the period 1984-1989. According to its mandate, UNDRO covers all aspects of disaster relief as well as preparedness and prevention activities. It must act and be seen to act on behalf of the Secretary-General in discharging this mandate.

4. In relief mobilization and co-ordination, in addition to its involvement in African countries where emergency situations persisted through 1986 and 1987, UNDRO was called upon to act in a wide variety of other disaster situations, many of which caused extensive devastation and some of which were of a complex nature. A number of these situations were in part or entirely due to armed conflict. Special attention was given to relief needs unlikely to be covered locally. To this end, the Office acted in close consultations with governmental and non-governmental organizations both in the field and at Geneva. There was close interaction with resident representatives of the United Nations Development Programme (UNDP) who represent UNDRO in the affected countries.

5. Effective mobilization and co-ordination of international emergency relief depends primarily on obtaining and rapidly disseminating relevant information. To this end, UNDRO continues to place the utmost importance on strengthening its capacity to collect, evaluate and promptly disseminate such information. Particular effort was placed on timeliness and reliability of the situation reports (SITREPS) prepared by UNDRO for the international community on the basis of information provided by resident co-ordinators in the field and assessments communicated by Headquarters of concerned organizations of the United Nations system. To expand

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further its information base and its capacity to disseminate disaster-related information, UNDRO continued to update its profiles of disaster-prone countries and is in the process of developing an International Disaster Management Information Network (UNDRONET) (see sect. IV of the present report, para. 49).

6. Activities related to disaster preparedness and prevention were significantly higher than in the previous biennium, as indicated, inter alia, by the number of requests received from developing countries to strengthen their national emergency services. In pre-disaster planning, UNDRO provided advice and expertise on the utilization of early warning systems, the elaboration and implementation of disaster contingency plans, on institution building within governmental machinery to deal with disaster situations, on strengthening the capacity of national Governments to assess relief requirements and to elaborate plans for the monitoring and distribution of relief supplies in the post-disaster period.

7. Over the past two years, some 30 such projects, both national and regional, have been in various stages of implementation. In medium- and longer-term aspects of disaster mitigation, priority has been given to regional, interregional and global activities, such as the Seismic Risk Reduction Project in the Balkans, the Co-operative Project for Seismic Risk Reduction in the Mediterranean Region, the Pan Caribbean Disaster Preparedness and Prevention Project, and the Seminar on Earthquake Prediction and Mitigation of Earthquake Losses sponsored by UNDRO, the United Nations Development Programme (UNDP), the United Nations Educational, Scientific and Cultural Organization (UNESCO) and the Union of Soviet Socialist Republics.

8. For the biennium 1988-1989, the work and activities of UNDRO will be guided by resolutions adopted by the General Assembly. Guidance for action has already been provided in the report of the Secretary-General (A/42/657), endorsed by the General Assembly. In this connection, it is necessary to recall action already taken concerning the strengthening of links between UNDRO headquarters and the UNDP resident representatives, who represent UNDRO in the field. To this effect, as recommended in the Secretary-General's report, a joint UNDRO/UNDP Task Force was established to work out improved mechanisms for co-operation and has completed its work.

9. Specific examples of the activities of UNDRO are provided in the following sections of the present report: section II on disaster relief co-ordination; section III on disaster mitigation: a combination of prevention and preparedness; and section IV on disaster information. Financial and administrative information is presented in section V, and in tabular form in annexes I, II and III to the present report.

II. DISASTER RELIEF CO-ORDINATION

10. This subprogramme constitutes the core of UNDRO's activities and, as such, is accorded the highest priority in the allocation of the Office's human and financial resources. A brief outline of individual relief operations for major emergencies is presented in annex V to the present report.

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11. During the 1986-1987 biennium UNDRO was involved, to varying degrees, in 110 disaster situations, of which a considerable number were of a long-standing nature calling for UNDRO assistance over a lengthy period; this represented an increase of 11 per cent over the previous biennium. Major disasters which required the involvement of UNDRO included cyclonic storms and floods in Bangladesh, the release of toxic gas from Lake Nyos in Cameroon, an earthquake in El Salvador, floods in Bolivia, drought in Ethiopia, floods in Haiti, armed conflict and floods in Lebanon, a cyclone in Madagascar, drought/destabilization in Mozambique, floods in Peru, a cyclone in the Solomon Islands, drought in the Sudan, a typhoon in Viet Nam, a forest fire in China, an earthquake in Ecuador, a tidal wave in the Maldives, a cyclone in Vanuatu, civil strife in Democratic Yemen and a wide-spread locust threat in Africa.

12. These disaster situations drew wide attention and support from the world community, with UNDRO discharging its role of mobilizing and co-ordinating relief assistance, providing an independent assessment of damage and needs and keeping the donor community informed of developments. Joint inter-agency missions were fielded, as necessary, to make in-depth assessments of the situations and relief needs, and to set up concerted relief programmes; within this framework, bilateral donors, the United Nations system and non-governmental organizations provided assistance to the affected populations.

13. The increased number and scope of emergency situations calling for international attention stretched the Office's resources to the maximum. In this context, co-operation with the organizations of the United Nations system, as well as with the other intergovernmental and non-governmental organizations active in humanitarian relief, was further reinforced. Inter-agency consultations both at UNDRO headquarters and in the field were stepped up, and various multi-agency assessment and evaluation missions were organized. The locust infestation throughout Africa might be cited as an example of supportive co-operation on the part of UNDRO with the specialized agency substantively concerned. While the Food and Agriculture Organization of the United Nations (FAO) oversaw the situation and monitored its evolution, UNDRO kept the donor community continually informed of emerging needs through its global communications system, thereby helping in the mobilization of resources.

14. Also in connection with its function of mobilizing funds, UNDRO organized a meeting at Geneva on 31 March 1987 convened by the Secretary-General, on humanitarian assistance to Mozambique. The meeting, which was chaired by the United Nations Disaster Relief Co-ordinator, was attended by the Prime Minister of Mozambique and by representatives of 53 Governments, 2 intergovernmental organizations, 14 agencies of the United Nations system and 37 non-governmental organizations. A clear commitment towards alleviating the plight of the affected population in Mozambique was expressed by all participants. Relief contributions pledged at the meeting for 1987 amounted to approximately \$US 209 million from 19 donor Governments and one intergovernmental organization.

15. At the national level, through the assignment of field delegates, UNDRO was instrumental in strengthening local emergency management capacity under the responsibility of the resident co-ordinators, i.e., the UNDP/UNDRO resident

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representatives. Co-ordinating mechanisms were revised or established with representatives of bilateral donors, the United Nations system and other intergovernmental and non-governmental organizations in order to formulate and implement operational emergency programmes with the active participation of national authorities in the affected countries. Also data were continuously updated on the remaining relief needs for submission to the international donor community.

16. In addition to information, logistics and transport at the national level are critical to effective relief management. Consequently, the Office is frequently called upon to assist in monitoring transport facilities and in establishing distribution networks, as well as in the mobilization of resources for, and the organization of, large-scale shipping and air-drop operations, such as those carried out in Angola, Ethiopia and Mozambique. The importance, duration and complexity of these operations has often required the establishment of innovative monitoring, data collection and analysis systems in the countries concerned.

17. In this connection, UNDRO successfully launched special appeals, widely supported by donors, for funds to cover the transport costs of relief donations in kind. UNDRO also devised and introduced a new contribution reporting form, in which donors are asked, inter alia, to indicate their cash contributions to the transportation costs of the relief supplies they are providing.

18. As the donor community responds to an international appeal, the Office continuously records contributions and pledges and reconciles them against outstanding needs. This process continues throughout the entire emergency phase and enables the Office to promptly caution against an over-supply of certain relief items or the duplication of ongoing efforts. It also helps donors to adjust to rapidly changing conditions, to combine relief shipments and to avoid sending unsolicited relief items.

19. Once the relief needs have been effectively met, the Co-ordinator advances proposals in his final situation report for the rehabilitation and reconstruction period to follow. Specific needs for rehabilitation and reconstruction, often aimed at reinforcing the disaster preparedness capacity of the affected country, are developed in close consultation with the Government and concerned agencies of the United Nations system.

20. In the period under review, cash contributions channelled through UNDRO to cover concerted relief programmes and specific ad hoc requirements remained at the same high level as in the previous biennium. They were used to procure emergency supplies not otherwise provided to assist in overcoming bottle-necks in the transportation and distribution of relief items and to strengthen the proper handling and monitoring of aid arriving from abroad. In these programmes, UNDRO availed itself of the services of other agencies and non-governmental organizations operative in the field. In Lebanon for instance, UNDRO received and managed \$US 884,791 and arranged the implementation of a number of health and social welfare programmes through international and local non-governmental organizations. In Ethiopia, UNDRO was entrusted with more than \$US 5 million for a series of airlift and air-drop operations in 1986/1987. In Haiti, the Office sponsored a

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number of urgently needed relief and recovery projects for a total of \$US 961,000. With the \$US 10,985,000 obtained for Mozambique, the Office financed several essential ventures in the areas of health, infrastructure and logistics in co-operation with the Government of Mozambique, other United Nations agencies, and non-governmental organizations. Given that administrative procedures in donor countries frequently cause substantial delays between pledges and actual cash deposit, UNDRO maintains a revolving fund to advance aid between commitment and delivery.

21. An illustration of both the growing concern for the establishment and/or the strengthening of links between emergency assistance and long-term programmes aimed at finding durable solutions for the affected people and the problems pertaining to displaced persons, was the inter-agency mission led by the Office at the request of the Governments concerned. The inter-agency mission (UNDRO), the World Food Programme (WFP), FAO and the International Fund for Agricultural Development (IFAD) visited Algeria and Mali in December 1986, in order to assess immediate needs of displaced people from Mali and Niger in the camps of the Algerian Sahara, and to examine the prospects for the settlement of the displaced persons from Mali in their own country. Following this mission, programmes to promote the return of the displaced persons were initiated in Mali by the agencies concerned, while emergency assistance was provided in the Algerian camps.

22. The capacity of the Office to respond to emergency needs was greatly enhanced by the full scale use of the UNDRO warehouse at Pisa, which was made possible by the support of the main donor, the Government of Italy, as well as other donors. The warehouse, which was constructed at the site of the United Nations Supply Depot at Pisa, became operational in early 1986 and accommodated a wide range of standard relief goods in sectors such as shelter, logistics equipment and water supply. Its location at the Pisa airport, which can accommodate any type of aircraft, makes it suitable for dispatching relief supplies to disaster-stricken developing countries at short notice. The operations of the warehouse, operating as a complement with UNIPAC (UNICEF Packing and Assembly Centre) at Copenhagen, were expanded considerably, following the receipt of donations from the Governments of Norway and Finland. Thanks to arrangements with these donors, the warehouse was replenished immediately after the dispatch of goods to disaster-stricken areas. At the end of 1987, consultations with the Government of Japan were at an advanced stage regarding its donation of emergency relief supplies for storage in the warehouse and their release to disaster-stricken areas in Africa and the Mediterranean region.

23. Owing to current United Nations financial constraints and corresponding cuts in the regular budget allocation for emergency cash grants, from \$360,000 to \$180,000 per year, the Office's capacity to respond to requests from disaster-affected countries for such grants was restricted. While 32 requests for emergency cash grants were received in the 1986-1987 biennium (see annex I to the present report), UNDRO was able to respond positively to only half of them, with an average of as little as \$22,000 per disaster (see annex IV).

24. As international relief operations involve numerous partners - the Government of the stricken country, bilateral donors, non-governmental organizations and the

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United Nations system - UNDR0 has continued to promote the establishment of consultative mechanisms at the local and international levels. It is through such mechanisms that concerted relief programmes are developed, priorities established, major pledges made known, responsibilities assigned to or divided among several interested donors, and their effective implementation ensured.

25. To reinforce co-operation with the National Emergency Relief Services for foreign disaster assistance, and to make maximum use of their collective experience and advice, meetings of heads of these services are now organized on a periodic basis. The first such meeting was held at Geneva from 18 to 20 June 1986 with government officials in charge of some 20 services responsible for mobilizing and co-ordinating their country's response to appeals for international assistance. The meeting examined the successes and weaknesses of a number of recent international relief operations, ways to improve the immediate assessment of damage and to identify emergency needs as well as to develop and carry out concerted relief programmes. In this connection, the officials of the National Emergency Relief Services reviewed the services provided by UNDR0 in disaster relief, preparedness and prevention, and put forward a number of proposals designed to strengthen co-ordinating mechanisms both at Headquarters and in the field. The second meeting took place at Geneva from 6 to 10 April 1987. This meeting was enlarged to allow for the participation of the heads of the emergency services of specialized agencies and other organizations of the United Nations system as well as of the humanitarian non-governmental organizations. Main questions of common concern were: how to discourage the continuing practice of sending unsolicited relief supplies; how to avoid diverging assessments of unmet relief needs and the delivery of large-scale assistance into often inaccessible or insecure areas; and the conditions of deployment, utilization and training of foreign search and rescue teams. Both meetings called for the strengthening of the co-ordination and mobilization functions of UNDR0, and urged the Office to assume a more commanding role in exercising leadership and co-ordination. The Office was further urged to examine the feasibility of establishing regional offices. The third meeting of the National Emergency Relief Services is scheduled for early 1989. 1/

26. A post-operation evaluation procedure has been developed in consultation with officials of the National Emergency Relief Services to evaluate the impact of international relief activities and the role played by the Office. All parties involved in a relief operation are requested to participate in this evaluation by providing comments and data.

27. Some of the specific problem areas so far identified through this evaluation process follow:

(a) Programme of operations - other agencies. Some agencies do not automatically inform UNDR0 of their relief programmes in countries affected by a disaster. This important shortcoming is being highlighted in the regular inter-agency consultations of UNDR0;

(b) Circulation of evaluation reports on relief activities. Agencies that have undertaken evaluations of their own operations in disaster-affected countries do not as a rule share these with other partners in the operation. UNDR0 has been

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promoting the idea of circulating individual evaluation reports on relief activities in disaster-affected countries among the United Nations agencies, intergovernmental organizations and non-governmental organizations concerned, as a means of establishing an overall evaluation of co-ordinated programmes;

(c) Programmes and activities of non-governmental organizations. The exchange of information and reports on non-governmental organization relief programmes and activities in disaster-stricken countries has to be further stepped up to enable the Office to play its focal point role in providing information essential for the co-ordination of disaster relief;

(d) Delays in launching appeals for international assistance. At times, requests for international assistance following a disaster are slow in coming because of the reluctance of local authorities to admit their difficulties in handling the situation. By maintaining continuous contact with Governments through its resident representative, and effectively monitoring the situation from its inception, the Office is attempting to reduce the number of such cases;

(e) Unsolicited relief supplies. The provision of unsolicited relief items resulting from erroneous information and a desire to express solidarity with the affected country frequently tax its limited transport and distribution capacities. The Office is therefore advising donors to husband their resources after a disaster strikes until the exact requirements are assessed and made known;

(f) Designated focal point. In some instances, the absence of a designated focal point for disasters in the national administration hinders prompt and effective relief operations and co-ordination. The importance for Governments to assign this function clearly is therefore being strongly emphasized;

(g) Communication problems. In the aftermath of some major disasters, serious communication problems that hampered the timely flow of essential information have been experienced. By adding radio equipment and satellite facilities to its communications system, the Office hopes to overcome this problem in future;

(h) Delay in utilization of available relief funds. Unnecessary long delays sometimes occur between the time a cash contribution is made and the time the affected population benefits from it. Among the factors causing these delays are the late identification of priority requirements, uncertainty as to whether they can be procured locally, and lack of resources to ensure transportation and distribution. Apart from contributing to shortening these delays by providing advances from the Revolving Fund, the Office is endeavouring to overcome the above factors in co-operation with its representatives in the field.

III. DISASTER MITIGATION: A COMBINATION OF PREVENTION AND PREPAREDNESS

28. Disaster mitigation programmes are based on two principal components: firstly, prevention, to determine the scientific and technical nature of probable

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disaster-causing events that may occur in the future, and the means to protect people and property from their impact; and, secondly, preparedness, to determine primarily how people can be removed to safety and looked after when a disaster is imminent, or has occurred. In any one project, be it national or regional, the normal sequence is to start with prevention and proceed to preparedness activities, i.e., to base logistical plans on the study and understanding of probable future disaster scenarios. For instance, better siting and reinforcement of structures (a typical prevention measure) will simplify emergency management planning requirements, namely, preparedness. In view of the close interrelationship of prevention and preparedness, these activities are presented jointly under the heading of disaster mitigation, even though, for the purposes of programming they continue to be considered under two subprogrammes.

29. After the occurrence of a major disaster or emergency, attention is invariably drawn to the need for better pre-disaster planning as a prerequisite to efficient relief management. This involves a sequence of prevention and preparedness activities in order to determine:

(a) What types and magnitudes of disaster are likely to happen in the future and where;

(b) What will be their impact on people and property;

(c) What protective measures or emergency plans are required; how and by whom will they be prepared;

(d) By whom and using what criteria will the decision be taken to implement these measures;

(e) What post-disaster measures will be needed for the survivors whether in situ or displaced.

30. The response to these questions requires the technical knowledge and co-operation of various disciplines, including engineering, planning, civil defence, public safety, health, welfare and public information.

31. The role of UNDRO is to encourage disaster-prone countries to address the above questions systematically in the high-risk areas, and to bring together the various disciplines involved in finding the most appropriate methods of responding. Each type of disaster and each high-risk area require a specific combination of prevention and preparedness measures. The growing number of requests for assistance in disaster mitigation addressed to the Office reflects the increasing understanding that by committing modest resources to such activities it is often possible to reduce loss of life by half and property losses by not less than 5 to 10 per cent. The procedure followed by UNDRO is to provide staff who advise and assist the national authorities, firstly, to identify the most appropriate procedures for disaster mitigation, secondly, to develop a draft work programme and calendar, and thirdly, to identify sources of funding, including UNDP/IPF (indicative planning figure) or special project grants by donor countries. For disasters that affect large areas, transgressing national boundaries, a

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regional approach is often called for. The following are some examples of such regional and national approaches.

Asia

32. In the Asian region, the focal point for the disaster mitigation activities of UNDRO is the Asian Disaster Preparedness Centre, established in January 1986 at the Asian Institute of Technology at Bangkok. The Centre, with a full-time staff of 11, responds to the specific needs of countries in the region, providing services in training, information management, pre-disaster planning, the application of appropriate technology and public awareness programmes. The Centre runs courses in disaster management for national officials responsible for disaster co-ordination. Since its inception, the Centre has catered to the needs of the following countries in the region: China, India, Indonesia, Papua New Guinea, the Philippines, Sri Lanka, Thailand and Viet Nam, as well as the Territory of Hong Kong.

The Pacific

33. Towards the end of 1987, UNDRO began to lay the groundwork for a disaster mitigation programme for the Pacific region. Discussions were held with government representatives, the South Pacific Bureau for Economic Co-operation (SPEC) and the regional UNDP Office at Fiji. The main objective of the programme is the strengthening of the capability of the Pacific islands to cope with disasters, particularly those caused by cyclones. Activities envisaged include short-term training exercises in disaster management, both within the countries and abroad for national officials and updating of national disaster preparedness plans, emergency communication and community preparedness. It is planned to establish the programme in the Fijis and to collaborate closely with the regional tropical cyclone project of the World Meteorological Organization (WMO) based at Nadi, as well as with organizations providing bilateral disaster preparedness assistance in the region.

The Caribbean

34. The Pan-Caribbean Disaster Preparedness and Prevention Project (PCDPPP), which is based in Antigua and serves 29 participating States and Territories, has been consolidated by securing direct contributions from the participating countries in its management and financing. The project is implemented in collaboration with the Pan American Health Organization/World Health Organization (PAHO/WHO) and the League of Red Cross and Red Crescent Societies. Since 1981 a variety of initiatives has been launched which resulted in an increased capacity for disaster management at the national and regional levels. These include training of personnel, technical assistance, surveys of vulnerable areas, the creation and activation of national disaster preparedness offices and the improvement of the emergency telecommunications systems. The project's activities for 1986/1987 focused on hazard mapping programmes in risk areas, the improvement of resistance of buildings to earthquakes and hurricanes by training building contractors and inspectors, the implementation of public awareness measures and the setting up of a regional disaster information centre. The programme also includes technical assistance for the development of national emergency policies and contingency

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plans, training in hurricane shelter procedures and damage assessment and simulation exercises. The project serves as a regional pool of expertise in the Caribbean and as a focal point for the sharing of experiences, promotion and co-ordination of efforts undertaken in the disaster mitigation field. To this effect, it maintains linkages with relevant international organizations (UNDP, the United Nations Environment Programme (UNEP), UNESCO, WHO and the International Telecommunication Union (ITU)), and regional institutions (University of the West Indies, Caribbean Council of Engineering Organizations, Caribbean News Agency, Seismic Research Unit, and others). Funding for PCDPPP was provided by the Governments of Canada, the Netherlands and the United States of America, as well as by the participating States and Territories and UNDRO. It is planned to extend the project until the end of 1990 when the participating States and Territories have agreed to assume the project's activities by establishing a self-sustained regional body.

Latin America

35. In Latin America, the recent volcanic eruptions in Colombia (1985) and Guatemala (1987), the earthquake in San Salvador (1986), earthquakes and landslides in Ecuador (1987), and floods in several countries have heightened the awareness of Governments that mitigation programmes can effectively reduce the impact of disasters in human and economic terms. In response to requests from several countries, a regional disaster prevention and preparedness programme was initiated in 1986, with activities focused initially on earthquake, volcano and tsunami hazards. The main activities have included a workshop and field training course in volcanic hazard mapping, attended by 18 participants from nine countries, and a similar workshop and training course on the recognition and mapping of active earthquake zones. The preparation of reliable hazard zoning maps is an essential prerequisite in developing emergency plans, but one which, so far, has been taken for only a few high-risk areas. Initially the programme will focus on Andean countries and will later encompass other countries of the continent. Financing is principally provided by the Office of Foreign Disaster Assistance of the United States Agency for International Development (USAID).

The Mediterranean

36. Following a series of devastating earthquakes in the Mediterranean region over the last 10 years, and repeated calls by the countries affected for concerted action to reduce earthquake losses, preparatory work has been carried out to develop a regional co-operative project on seismic risk reduction. After surveying country needs, a project has been elaborated by UNDP and UNDRO in co-operation with the United Nations Centre for Human Settlements (UNCHS), UNEP, UNESCO and the United Nations Industrial Development Organization (UNIDO) to assist the Mediterranean countries in developing and sharing innovative techniques for seismic risk assessment, mitigation and emergency planning. The project is largely based on experience of earlier seismic risk reduction projects in the Balkan region, and is funded mainly by a contribution from Italy.

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National projects

37. In addition to these regional projects, UNDRO has co-ordinated single-country activities in Egypt, Haiti, Indonesia, Madagascar, Nepal and Niger, and organized or contributed to various single or multi-country disaster mitigation seminars and training activities. Examples of such projects follow.

Egypt

38. Egypt has identified flash flood and earthquake as two disaster types for which it wishes to strengthen its prevention and preparedness capacity. UNDRO has initiated and is monitoring, in close liaison with UNDP, a technical assistance project. A preparatory assistance mission was fielded to review hazards, assess national disaster mitigation and management capacities, and identify future evaluation planning and training needs.

Indonesia

39. A project to strengthen all aspects of disaster management and mitigation has been implemented since mid-1986 with the national authorities, UNDP and USAID. The project involving four ministries has four main areas of activity: training; organization and procedures; information flow; and risk monitoring. It has identified disaster mitigation options for inclusion in Indonesia's next five-year development plan. Nearly 200 national and provincial disaster management officials have already received training.

Madagascar

40. Madagascar has suffered major recent losses from cyclones and floods. In response to government requests for technical assistance to mitigate these, UNDRO has since 1985 supervised the execution of a UNDP-funded project for the establishment of a national network of emergency centres with related training and communication systems for disaster management. By the end of 1987 the project was almost completed.

Training seminars

41. In addition to various training activities undertaken as components of regional projects, several independent seminars have been held. In 1986, a Seminar on Earthquake Prediction and Mitigation was held at Dushanbe, USSR, and organized in co-operation with institutions of the host country as well as with UNDP and UNESCO. The purpose was to acquaint responsible officials from developing countries with the latest methods of earthquake risk reduction and disaster management, and to exchange experience and information on these.

Manuals

42. In response to the need expressed in many developing countries for simple interdisciplinary techniques to assess the risks of natural disasters and to design appropriate counter-measures, UNDRO is preparing a manual on disaster mitigation

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aspects of physical planning and building. The manual gives practical guidance to professionals and administrators at the regional and local levels, and will be a further addition to the existing UNDRO series entitled Disaster Prevention and Mitigation - A Compendium of Current Knowledge.

43. Another manual in preparation deals with mudflows, which have been prominent among the causes of recent disasters. This manual reviews current knowledge of the characteristics of mudflows, and techniques for the mitigation of their impact.

International Decade for Natural Disaster Reduction

44. With a view to providing concerted international action to reduce the human and material losses caused by national disasters, through the wider diffusion and application of scientific and technical knowledge, the idea of an international decade, first suggested by the President of the United States Academy of Sciences in his address to the World Conference on Earthquake Engineering in July 1984, has been welcomed and supported by professional communities throughout the world. Subsequently, the General Assembly, at its forty-second session, adopted resolution 42/169 on 11 December 1987, designating the 1990s as the International Decade for Natural Disaster Reduction. Following the establishment by the Secretary-General of a United Nations Steering Committee for the Decade, UNDRO acted as host in April 1988 to the first meeting of an Inter-Agency Working Group at which the relevant activities of the organizations of the United Nations system were reviewed and recommendations were made to the Steering Committee concerning action to be taken in preparation for the launching of the Decade in January 1990. Steps are now being taken with a view to preparing the reports that the Secretary-General has been requested to submit to the General Assembly at its forty-third and forty-fourth sessions. These include:

- (a) The constitution of an International Preparatory Group of Experts to assist the Secretary-General in the planning and preparation of the Decade and to advise on its initial programme content;
- (b) Meetings of an Inter-Agency Steering Committee and of an Inter-Agency Working Group, to define the role of the United Nations system in promoting the Decade and in supporting national and international activities to be undertaken as part of the Decade programme;
- (c) Consultations with interested international non-governmental organizations, with a view to determining the role that these organizations may play in the planning and implementation of the Decade's programme of activities.

IV. DISASTER INFORMATION

45. Requests to UNDRO for disaster-related information have increased, especially from national disaster offices, training institutes and technical services and from the field offices of various international and non-governmental organizations. There has also been a marked increase in the number of visits by disaster specialists and researchers to consult the UNDRO Reference Centre and Data Bank.

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46. Several new volumes on disaster mitigation and management have been published, which include the French and Spanish editions of volume 12 of the UNDRO series Disaster Prevention and Mitigation - A Compendium of Current Knowledge, entitled Social and Sociological Aspects 2/ and the Spanish edition of the Manual on Volcanic Emergency Management. 3/ In co-operation with WHO/PAHO, UNDRO has ensured the publication of a French edition of five manuals on the health management aspects of disasters.

47. A bilingual (English and French) directory of national and international schools, institutions and agencies providing training in disaster management, 4/ and a list of national officials responsible for the management of disasters and other emergencies, emergency plans and disaster legislation 5/ have also been published. As at 1 January 1987, a computerized listing of disasters by types of event and location, with a brief summary of human and economic losses, was established and will be further developed.

48. The data processing and communications activities of UNDRO again showed a considerable expansion during 1986-1987. In the area of data processing, one major new system was introduced, the UNDRO International Disaster Management Information Network - UNDRONET (see description below) and two new data bases were designed and recorded. In addition, significant enhancements to existing components were made, including a complete redesign of the contributions recording system, whose implementation was completed during the reporting period. In the area of telecommunications, the UNDRO computerized system for distribution of disaster situation reports (SITREPS) was also improved significantly, allowing for the efficient, cost-effective distribution of these SITREPS to an important new body of potential donors in North America. As a result, the total number of SITREP copies distributed to potential donors around the world increased from some 50,000 in the 1984-1985 biennium to approximately 65,000 in 1987-1988. The Office also transmitted in the reporting period a total of 72,378 telexes, including copies of the 65,000 SITREPS.

49. A new information network for international disaster management called UNDRONET has been launched. This is a network by which members of the world-wide disaster management community can have direct contact with each other, either through a computer terminal or through a micro-computer, and which provides them instantaneously with both background and operational disaster-related information. UNDRONET provides an electronic mail service, a bulletin board, and access to data bases which can be searched for specific disaster-related information and is connected to all other parallel United Nations networks. It is expected to bring about an important reduction in UNDRO telecommunication costs. Table 1 below shows the magnitude of the present expenditures and consequently the potential for reducing these by the introduction of UNDRONET.

50. New systems for the distribution of multiple-address telexes for the distribution of UNDRO situation reports and for contributions reporting have been introduced. Remote computer-to-computer communications have been expanded by the participation of UNDRO in the EARN/BITNET information network, as well as by the installation of micro-computers in disaster-affected countries.

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Table 1
Statistics on disasters and telecommunications

	<u>Biennium</u> <u>1984-1985</u>	1986	1987	<u>Biennium</u> <u>1986-1987</u>
Disasters with UNDRO involvement	98	52	58	110
Situation reports issued	335	174	151	325
Number of telex messages transmitted	56 327	36 253	36 125	72 378
(of which, SITREPS	50 000	32 000	33 000	65 000)
Costs (\$US)	180 963	150 162	162 004	312 166
Transmission time (hours)	7 165	4 990	4 635	9 625
Costs per hour (\$US)		30 1	34 95	32 43

V. ADMINISTRATION AND FINANCE

51. The present organizational structure of UNDRO is set out in document ST/SGB/Organization, section N/Rev.2 and Amend.1 of 1984, and responds to proposals made by the Administrative Management Service of the Secretariat in 1981.

52. The staffing table of UNDRO for 1986/1987 appears in annex II to the present report. A total of 37 posts are financed from the regular budget and 19 from the sub-account of the UNDRO Trust Fund for the strengthening of the Office.

53. For the biennium 1986-1987, the General Assembly appropriated an amount of \$US 6,418,300. Annex I to the present report shows the regular budgets for the three bienniums 1982-1983, 1984-1985 and 1986-1987, together with the evolution of appropriations and corresponding expenditures.

54. In so far as extrabudgetary resources are concerned, all voluntary contributions to UNDRO are credited to one of the accounts below:

(a) United Nations Trust Fund for General Disaster Relief, created by virtue of General Assembly resolution 2816 (XXVI) of 14 December 1971:

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<u>Biennium</u>	<u>Available balance at end of period</u>
	(United States dollars)
1982-1983	783 200
1984-1985	970 900
1986-1987	731 371

(b) UNDRO Trust Fund, sub-account for the strengthening of UNDRO, created by virtue of General Assembly resolution 3243 (XXIX) of 29 November 1974:

<u>Biennium</u>	<u>Opening balance</u>	<u>Income</u>	<u>Expenditure</u>	<u>Closing balance</u>	<u>Combined allotments for the two years</u>
					(United States dollars)
1982-1983	923 606	1 254 446	1 395 011	783 041	1 532 800
1984-1985	783 041	1 729 019	1 505 412	1 006 648	1 992 400
1986-1987	1 006 648	2 316 462	2 188 308	1 134 802	2 003 800

(c) UNDRO Trust Fund, sub-account for technical co-operation created by virtue of General Assembly resolution 3532 (XXX) of 17 December 1975:

<u>Biennium</u>	<u>Opening balance</u>	<u>Adjustment to prior period</u>	<u>Income</u>	<u>Expenditure</u>	<u>Closing balance</u>
					(United States dollars)
1982-1983	573 017	-	702 686	1 127 233	506 368
1984-1985	506 368	-	1 617 693	1 826 368	1 019 145
1986-1987	1 019 145	150 000	9 987 984	6 024 010	4 833 119

This sub-account includes income and expenditures for two separate entities:

- (i) The Pan-Caribbean Disaster Preparedness and Prevention Project for which the 1986/1987 closing balance represents \$US 312,233;
- (ii) The United Nations Supply Depot at Pisa for which the 1986/1987 closing balance represents \$US 3,369,888.

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As a result, the unearmarked portion available for technical co-operation activities as at 1 January 1988 stands at \$US 1,150,998.

(d) UNDRO Trust Fund, sub-account for emergency relief in specific countries (earmarked contributions channelled through UNDRO):

<u>Biennium</u>	<u>Income from Governments and intergovernmental organizations</u>
	(United States dollars)
1982-1983	15 103 360
1984-1985	15 993 065
1986-1987	15 331 671

55. The sub-account for unearmarked emergency assistance to supplement regular budget grants, established by General Assembly resolution 3532 (XXX), has been inoperative during the period under review and continues to have available funds of \$US 10,000 by the end of 1987.

56. As can be seen in paragraph 54 (b) above, the opening balance as at 1 January 1988 of the UNDRO Trust Fund sub-account for the strengthening of UNDRO is still encouraging and amounts to \$US 1,134,802 compared to the 1 January 1987 opening balance of \$US 1,006,648. However, it should be recalled that more than one third of UNDRO's staffing and associate costs (i.e., travel and communication) are met by voluntary contributions. UNDRO therefore has the vast responsibility of securing the financing for this sub-account each year. Moreover, any demand for an increase in the level of activity entails expenditures in excess of the level of the Fund.

Notes

1/ See documents UNDRO/86/04 and UNDRO/87/5.

2/ Office of the United Nations Disaster Relief Co-ordinator, Geneva, 1986 (GE.85-02838).

3/ Ibid., GE.85-02770.

4/ Ibid., GE.86-03149.

5/ Ibid., GE.87-01496.

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Regular budget: comparisons of the budget appropriations and expenses for the Office of the United Nations Disaster Relief Co-ordinator (including New York Liaison Office) during the bienniums 1982-1983, 1984-1985 and 1986-1987, by object of expenditure

(Thousands of United States dollars)

Object of expenditure	1982-1983		1984-1985		1986-1987		Percentage of growth between 1982-1983 and 1986-1987	
	Approp.	Expenses	Approp.	Expenses	Approp.*	Expenses	Appropriation	Expenses
Salaries								
Established posts	2 923.2	2 822.7	2 861.5	2 911.5	3 882.4**	4 114.3**	32	45
General temporary assistance	127.7	103.7	91.5	91.8	133.6	91.5	4	(11)
Consultants	12.6	12.6	10.4	10.1	17.0	8.9	34	(29)
Ad hoc expert groups	-	-	18.1	-	20.8	12.0	100	100
Overtime	10.9	6.5	10.1	10.4	15.1	2.6	38	(60)
Common staff costs								
Representation allowances	7.9	7.9	9.2	9.2	9.2	4.9	16	(37)
Other common staff costs	944.5	945.1	919.8	918.3	1 173.4**	1 085.1**	24	14
Travel of staff	247.2	246.6	229.8	228.5	293.1	232.0	18	(5)
External printing and binding	4.6	4.6	-	-	12.3	-	167	(100)
General operating expenses and communications	88.2	87.0	71.8	72.0	103.6	124.7	17	43
Hospitality	1.2	1.0	1.0	0.9	2.0	0.6	66	(40)
Data processing equipment	-	-	-	-	35.8	17.9	100	100
Grants	960.0	960.0	953.6	943.6	720.0	357.0	(25)	(62)
TOTAL	5 328.0	5 197.7	5 176.8	5 196.3	6 418.3	6 051.5	20	16

* Should be read in conjunction with General Assembly resolution 42/213 of 21 December 1987 and the report of the Fifth Committee.

** Includes one P-5 temporary post.

ANNEX II

Staffing table of the Office of the United Nations
Disaster Relief Co-ordinator

	Regular budget 1986-1987	Extrabudgetary resources 1986-1987	Total
Professional category and above			
USG	1	-	1
ASG	-	-	-
D-2	1	-	1
D-1	3	-	3
P-5	6	2	8
P-4	6	1	7
P-3	3	2	5
P-2/P-1	3	-	3
Total	23	5	28
General Service category			
Principal level	3	-	3
Other levels	11	14	25
Total	14	14	28
Grand total	37	19	56

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TRUST FUNDS OF THE UNITED NATIONS DISASTER RELIEF CO-ORDINATOR

Statements of income, expenditure, assets and liabilities as at 31 December 1987 for the biennium 1986-1987

	Disaster relief assistance	Strengthening of UNDRD	UNDRD Pan Caribbean preparedness and prevention project	UNDRD/UNEP projects	UNDRD emergency relief assistance	Disaster prevention and pre-disaster planning*	Total
I. Statement of income and expenditure for the biennium 1986-1987 as at 31 December 1987 (expressed in \$US)							
Income							
Pledged contributions	12 331 559	1 434 789	1 232 137	-	-	7 237 978	22 236 463
Public donations	633 913	10 103	-	-	-	-	644 016
Subventions	1 366 199	410 995	22 000	-	-	742 514	2 541 708
Interest income	679 488	132 973	62 762	9 864	-	233 082	1 118 169
Miscellaneous income	33 384	327 602	21 262	-	-	436 247	818 495
Total income	15 044 543	2 316 462	1 333 161	9 864	-	8 649 821	27 358 851
Expenditure							
Staff and other personnel costs	582 241	1 904 402	535 773	-	-	544 289	3 566 705
Travel	112 853	16 230	166 167	-	-	182 024	477 274
Contractual services	13 520	750	14 356	35 621	-	12 829	77 076
Operating expenses	1 160 846	266 926	112 936	15 553	-	334 357	1 890 618
Acquisitions	1 246 232	-	55 205	-	-	355 809	1 657 246
Fellowships, grants, other	12 058 639	-	378 600	-	-	3 331 664	15 768 903
Total expenditure	15 174 331	2 188 308	1 263 037	51 174	-	4 760 972	23 437 822
Excess of income over expenditure	(129 788)	128 154	75 124	(41 310)	-	3 888 849	3 921 029
II. Statement of assets and liabilities as at 31 December 1987 (expressed in \$US)							
Assets							
Cash	7 636 356	963 565	545 730	-	-	7 021 864	16 167 515
Accounts receivable	1 269 209	-	104 775	-	-	118 244	1 492 228
Operating funds provided to executing agencies	1 393 900	-	-	-	-	-	1 393 900
Receivable from United Nations General Fund	-	184 063	-	37 371	10 000	1 846	818 495
Total assets	10 299 465	1 147 628	650 505	37 371	10 000	7 141 954	19 286 923
Liabilities							
Accounts payable	1 036 922	-	10 919	-	-	2 621 069	3 668 910
Unliquidated obligations	2 498 839	12 826	69 706	-	-	-	2 581 371
Deferred income	230 000	-	-	-	-	-	230 000
Due to United Nations General Fund	254 848	-	257 647	-	-	-	512 495
Total liabilities	4 020 609	12 826	338 272	-	-	2 621 069	6 992 776
Fund balance							
Balance available at 1 January 1986	6 528 290	1 006 648	237 109	78 681	10 000	782 035	8 642 763
Adjustments to opening balance	(119 646)	-	-	-	-	(150 000)	(269 646)
Excess of income over expenditure	(129 788)	128 154	75 124	(41 310)	-	3 888 849	3 921 029
Balance available at 31 December 1987	6 278 856	1 134 802	312 233	37 371	10 000	4 520 885	12 294 147
Total liabilities and fund balance	10 299 465	1 147 628	650 505	37 371	10 000	7 141 954	19 286 923

* Including the UNDRD/PISA supply depot.

ANNEX IV

Disaster involvement of the Office of the United Nations
Disaster Relief Co-ordinator - 1986/1987

Statement of UNDRO grants as at 31 December 1987

<u>Date</u>	<u>Country</u>	<u>Situation</u>	<u>Amount in \$US</u>
16 January 1986	Bolivia	Floods	30 000
3 February 1986	Democratic Yemen	Civil strife	2 000
17 February 1986	Lebanon	Armed conflict	50 000
25 March 1986	Peru	Floods	25 000
5 September 1986	Haiti	Fire and floods	30 000
19 November 1986	El Salvador	Earthquake	30 000
7 January 1987	Fiji	Cyclone Rajah	20 000
9 March 1987	Vanuatu	Cyclone	10 000
9 March 1987	Chad	Displaced persons	30 000
27 April 1987	Maldives	Floods	10 000
28 April 1987	Ecuador	Earthquake	25 000
25 May 1987	China	Forest fire	30 000
28 August 1987	Bangladesh	Floods	25 000
8 October 1987	Guatemala	Floods	10 000
20 October 1987	Lesotho	Snowstorm	10 000
31 December 1987	Viet Nam	Typhoon "Maury"	20 000
Total as at 31 December 1987			<u>357 000</u>

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ANNEX V

The response of the Office of the United Nations Disaster Relief
Co-ordinator to disaster situations: brief description of
relief operations

Locust infestations - Africa

1. The return of the rains after years of drought created ideal conditions for the reproduction of locusts and the development of infestations over wide areas in the middle of 1986. For the first time in half a century, all four species commonly found in Africa simultaneously invaded millions of hectares of cropland in approximately 20 countries of eastern, central and southern Africa.
2. After the Food and Agriculture Organization of the United Nations (FAO) had issued its first appeal in May 1986, UNDRO co-operated with the FAO Emergency Centre for Locust Operations (ECLO) in strengthening the mobilization and co-ordination of resources, recording emergency needs and contributions and in reporting to the international community on the development of the situation.
3. During the second half of 1986, the Office issued 11 situation reports both in English and French. A further 10 situation reports were issued in 1987.
4. To speed up the transfer of information, a computer-to-computer link allowing direct contact was established in September 1986 between the offices of FAO/ECLO at Rome and UNDRO at Geneva.
5. The Office also took a leading role in briefing non-governmental organizations interested in helping to combat the locust threat, by organizing special information meetings on the subject for their benefit. The meeting was attended by more than 40 non-governmental organizations, as well as by representatives of permanent missions and of organizations of the United Nations system.
6. At the recommendation of the United Nations Disaster Relief Co-ordinator, the Secretary-General issued a statement on 14 August 1986 drawing attention to the threat posed to food production in Africa by the locust infestations and pledging full United Nations support for the efforts of FAO to assist the affected countries. UNDRO/United Nations Development Programme (UNDP) resident representatives were also urged to reactivate the local emergency co-ordinating mechanisms which had functioned so successfully during the 1985 drought.
7. The sustained and effective collaboration between UNDRO and FAO drew a timely response from the donor community, and FAO was able to report that during the two years of the campaign as much assistance had been received as in the whole of the previous 20-year period.

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Bangladesh

8. A tropical depression hit the southern islands of Bangladesh on 5 June 1987, followed by a six-foot tidal wave which swept over low-lying islands. UNDRO monitored the situation and kept the international community informed through a number of information reports.

9. As heavy rains followed, the situation further deteriorated, and on 17 August the Government of Bangladesh requested international assistance. The disaster killed 682 persons, affected an estimated 23 million people, and damaged or destroyed some 780,000 houses and 1,100 miles of embankments. Thousands of cattle perished and over 4 million acres of standing crops were either totally or partially destroyed. From 19 August until the end of October, when the situation began to ease, UNDRO issued nine situation reports. Donor response to the appeal, with contributions in cash and kind, exceeded \$US 60 million reported to UNDRO.

10. The Disaster Relief Co-ordinator visited Bangladesh from 1 to 5 September at the invitation of the Government for talks with senior officials, and later reported to the Secretary-General.

11. In addition, UNDRO assigned two consultants to assist the UNDRO/UNDP Resident Co-ordinator and the Government of Bangladesh in assessing the damage, in identifying unmet needs and in drawing up rehabilitation programmes.

Bolivia

12. During the first quarter of 1986, heavy rains in the areas of Cochabamba, Santa Cruz and Titicaca in Bolivia led to widespread floods which affected 250,000 people and damaged or destroyed 10,000 homes and 44,000 hectares of crop land.

13. Emergency needs were clearly identified, and contributions of close to \$US 3 million were reported. UNDRO itself made a cash grant of \$US 30,000, which was used for the local purchase of tents and tools. The United Kingdom of Great Britain and Northern Ireland channelled a total of \$US 58,000 through UNDRO for the local purchase of relief supplies and the provision of medicines.

14. In March 1986, at the request of the UNDRO/UNDP Resident Representative, an UNDRO delegate was assigned to co-ordinate relief efforts. In addition, an UNDRO consultant spent three months in Bolivia (December 1986-February 1987) to assist the Civil Defence authorities to build up a co-ordinating structure to cope with the logistics of disaster relief.

15. A contribution of \$US 60,000 from the Government of Italy to UNDRO was used to finance two projects (run by non-governmental organizations and the UNDP/United Nations Centre for Human Settlements (UNCHS)) for resettling flood victims as well as to purchase communications equipment for the civil defence.

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Cameroon

16. An emission of toxic gas on 21 August 1986 from the volcanic Lake Nyos, in sparsely populated north-western Cameroon, caused the death of 1,500 people and led to the evacuation of 3,000 others. On 26 August, the Government of Cameroon requested UNDRO to launch an international appeal for certain basic relief supplies. By the following day, immediate emergency needs had been covered and UNDRO asked donors to delay the dispatch of further aid until confirmation of any unmet needs.

17. Probably owing to its unusual cause, the Lake Nyos disaster gave rise to extensive coverage by the mass media, which resulted in reports that were not always accurate. While the disaster affected only a limited number of people, reports show that international assistance far exceeded the actual needs.

China

18. In May 1987, as UNDRO was issuing its first information report on a forest fire in China, a major blaze had already begun to engulf towns and more than 300,000 hectares of forest in Heilongjiang Province. The fire spread rapidly due to strong winds and dry weather. Although thousands of fire-fighters, soldiers and civilians fought the flames on several fronts and the army carried out cloud-seeding, the fire could not be extinguished until one month later when widespread rain finally brought it under control.

19. Total deaths amounted to 193. The seriously injured numbered 226, and 56,000 people were rendered homeless. The fire devastated more than 1 million hectares, including 700,000 hectares of forest land, and damaged 12,000 houses. Losses were estimated at over \$US 110 million.

20. Detailed reports from the UNDRO/UNDP Resident Representative in China enabled the Office to keep the international community fully informed of the relief needs. Donors provided fire extinguishers, food and other relief materials.

21. The UNDRO emergency grant of \$US 30,000, together with those of UNDP and the United Nations Children's Fund (UNICEF), was utilized for the purchase of transport equipment to assist the fire-fighting operations. Total contributions reported to UNDRO amounted to \$US 5.85 million.

Ecuador

22. A severe earthquake struck four north-eastern provinces of Ecuador during the night of 5 and 6 March 1987. Two major shocks measuring 6.0 and 6.8 on the Richter scale were followed by a series of aftershocks. Over 1,000 people were killed and 26,000 rendered homeless. Approximately 75,000 people living in the Amazon region were cut off from the rest of the country. The rupture of Ecuador's major oil pipeline resulted in considerable loss of export earnings.

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23. At the request of the Government, an international appeal for emergency assistance was issued. Contributions in cash and kind from the international donor community amounted to some \$US 7.5 million, of which over \$US 300,000 were channelled through UNDRO. Damage to the country's social and economic infrastructure was estimated at \$US 185 million.

El Salvador

24. The earthquake that occurred on 10 October 1986 caused 1,200 deaths, injured more than 10,000 people and rendered some 50,000 families homeless. Between 10 October and 5 November 1986, UNDRO issued 14 situation reports and recorded contributions in cash and kind valued at \$US 12.5 million. Subsequent contributions for rehabilitation and reconstruction brought the total to over \$US 75 million.

25. Within 48 hours, donors were alerted to the fact that enough relief supplies covering immediate relief needs, as well as search and rescue teams, were being rushed to San Salvador by neighbouring countries, European donors, the United States of America, the European Economic Community (EEC) and the Pan-American Health Organization (PAHO). This permitted donors to adjust their contributions in kind to those items not yet covered or to reserve funds for the rehabilitation and reconstruction of San Salvador.

26. The French and Swiss rescue teams and an UNDRO delegate flew to San Salvador in a Swiss-chartered aircraft which left within 24 hours of the disaster. UNDRO arranged for the joint dispatch of medical supplies and other relief goods donated by four individual Governments, EEC, the International Committee of the Red Cross and several non-governmental organizations in another aircraft chartered by the Swiss Disaster Relief Unit.

27. UNDRO provided an emergency grant of \$US 30,000 towards re-establishing the drinking water supply in two hospitals of San Salvador.

Ethiopia

28. After the devastating drought of 1984/1985, the situation somewhat improved during 1986. In the course of that year, UNDRO organized a \$US 1 million air-drop operation which delivered within a period of two months 1,872 metric tons of emergency food in northern Wollo for distribution by the Relief and Rehabilitation Commission and World Vision International. This food was required to assist a still seriously famine-affected population of 135,000 people living in inaccessible areas around Sekota. This operation was undertaken with the Belgian Air Force which developed a unique low-altitude gravity extraction air-drop technique. a/

29. When the harvest was once again threatened by drought in the second half of 1987, a large-scale relief programme was drawn up in consultation with the Government and with the other humanitarian relief organizations operational in the country. An international appeal for 1.3 million metric tons of emergency food aid, as well as for large-scale assistance in the areas of logistics, agriculture, health and sanitation, was launched.

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30. Following a multi-donor relief transport capacity study, which indicated that surface transport alone was unlikely to meet emergency transport requirements in Eritrea and Tigray, a Transport Fund was established, out of which an air transport operation was financed. Some 12 donor countries responded to the emergency appeal of 13 November 1987 and pledged over \$US 9 million in record time. The UNDRO airlift became operational on 6 December and has continued uninterrupted to date.

31. During 1987, UNDRO issued 11 situation reports on the emergency and provided, through its delegates in the field, continued monitoring, information and emergency management functions to assist the Secretary-General's Special Representative.

Haiti

32. Following requests from the Government, which were endorsed by the Special Representative of the Secretary-General of the United Nations in Haiti, UNDRO launched appeals to the international community to fund several projects of an emergency nature. The projects were implemented under the supervision of an UNDRO delegate from the Swiss Disaster Unit, who was seconded to the "Office pre-desastre et de secours" (OPDES) - the national focal point for disaster preparedness and relief.

33. In June 1986, the south-western peninsula of Haiti was hit by serious flooding following exceptionally heavy rainfall over several days. Twenty thousand ha of agricultural land were devastated. Road infrastructure and irrigation systems were heavily damaged, several thousand houses destroyed, and 80,000 people affected. Several donor Governments and non-governmental organizations were quick to provide relief assistance. UNDRO developed a project to protect populated areas and agricultural land from the negative impact of future flooding. A total of \$US 690,000 were received in contributions from several Governments and a non-governmental organization. The project was implemented during the period from October 1986 to October 1987.

34. On 16 May 1986, a slum area of Port-au-Prince was destroyed by fire leaving its 2,400 inhabitants completely destitute. In collaboration with the disaster victims and the authorities concerned, UNDRO developed a project for the provision of basic housing undertaken by the disaster victims themselves with the help of architecture students from the university. An amount of \$US 130,000 received by UNDRO was used for the construction of a drainage channel, public water fountains and other sanitation measures. It was also decided to construct a water reservoir in case of future fire outbreaks.

35. On 10 July 1987, torrential rains hit Port-au-Prince and surrounding areas, causing important damage in some sectors of the capital and leaving 7,500 persons destitute. Following an appeal launched by the Government of Haiti through UNDRO, an emergency relief programme was implemented in co-operation with OPDES. An amount of \$US 130,000 was spent for local purchase of basic relief items and \$US 10,000 for equipment for the intervention brigades of OPDES. Contributions were received from five donors.

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Lebanon

36. The relief activity of UNDRO in favour of the affected population in Lebanon began on 11 June 1982 with the launching of the Secretary-General's appeal for relief assistance to alleviate the suffering resulting from the Israeli invasion. Since then, with the continuation of military activities in 1985, UNDRO involvement has been gradually increased. By October 1986 the value of assistance given or reported through UNDRO reached \$US 41,549,884. Because of increased emergency relief assistance and because of the evacuation of United Nations international staff, UNDRO made its representative at Beirut available to act also as Officer-in-Charge of the Office of the United Nations Co-ordinator of Assistance for the Reconstruction and Development of Lebanon as well as Chairman of the United Nations Co-ordinating Committee at Beirut.

37. In 1987, external military intervention in southern Lebanon was complicated by a further escalation in the so-called "camp war". The consequences of this day-after-day deterioration have been widespread impoverishment of a growing part of the population which is no longer able to satisfy its basic needs in health, food, housing and education.

38. As a response to the increased number of people in need, the Secretary-General launched a "special appeal" through UNDRO on 20 March 1987 for the provision of assistance to Lebanon. In close co-ordination with the Office of the Under-Secretary-General for Political and General Assembly Affairs, based on General Assembly resolution 41/196 of 8 December 1986, UNDRO continued monitoring the emergency situation in Lebanon. In March, September and December 1987, UNDRO organized information meetings for the representatives of Governments, United Nations agencies, intergovernmental and non-governmental organizations. Its representative at Beirut was joined by another delegate to co-ordinate relief activity in the country. The assistance in cash, materials and services given or reported through UNDRO in 1987 amounted to \$US 22,936,956 and provided help to Lebanese of all categories (displaced people, mothers and children, disabled, students, orphans, etc.) as well as people affected by natural disaster (floods).

39. Following a United Nations Inter-Agency Assessment Mission (4 to 10 October 1987), the Secretary-General appealed through UNDRO on 4 December 1987 for further emergency relief aid for a value of \$US 35,150,000, including food, health and sanitation facilities, education, shelter and fuel. So far, donors' response has exceeded \$US 6 million.

Mozambique

40. On 27 February 1987, UNDRO issued a special appeal by the Secretary-General for \$US 247 million in emergency assistance to the People's Republic of Mozambique. In March 1987, the Secretary-General convened a meeting at Geneva "to review the response to his appeal and to assess latest developments in regard to relief operations in Mozambique".

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41. Emergency relief contributions reported to UNDRO from 1 January to 31 December 1987 amounted to \$US 337 million, of which \$US 1,391,587 were channelled through UNDRO by the Governments of Australia, Canada, Cyprus, Finland, Ireland, Switzerland and the United Kingdom of Great Britain and Northern Ireland.

42. UNDRO provided an adviser and emergency liaison officer to support the United Nations Special Co-ordinator for Emergency Relief Operations in Mozambique (UNSCERO) in his tasks. The adviser assisted the Government in creating emergency mechanisms, including provincial emergency commissions in each of the affected provinces and the Emergency Operations Committee. In co-operation with the Swiss Disaster Relief Unit, the Office supplied the services of computer experts, computer equipment and training for UNSCERO and offices of the National Executive Commission for Emergency. Another adviser stationed at Maputo carried out field assessments and, in conjunction with the World Food Programme (WFP), identified logistics bottle-necks and advanced solutions.

43. In August 1987, the Office chartered a small aircraft to facilitate movements between Maputo and the provinces. The aircraft was used, inter alia, to allow UNSCERO, the Government of Mozambique and representatives of donor countries to monitor the evolution of the emergency situation and needs in otherwise inaccessible areas, as well as to undertake assessments of field projects by non-governmental organizations and United Nations agencies.

44. During the reporting period, UNDRO was directly involved in a number of projects, the main ones of which are listed below:

(a) Construction of a 275-meter-long cement bridge over the Incomati River at Moamba (north-west of Maputo) and repair of the Moamba-Sabie road;

(b) Rehabilitation of the Nhamatanda Hospital (Sufala) and a health care programme in Inhambane Province;

(c) Provision of seven trucks and radios to DPCCN (Department for the Prevention and Combat of Natural Disasters);

(d) Supply of 2,000 metric tons of aluminium sulphate for water treatment;

(e) Support for:

(i) The WFP airlift of food and supplies to seven isolated localities in the provinces of Sofala and Manica;

(ii) OXFAM/United Kingdom transport operations;

(iii) WFP-run coastal services;

(f) Local transportation of seeds in time for the sowing season;

(g) Supply of tents, generators, tools and galvanized corrugated iron sheets from the UNDRO warehouse at Pisa.

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Democratic Yemen

45. Fighting started in Aden on 13 January 1986. Following reports that the interim leader of Democratic Yemen requested international assistance, UNDRO launched an appeal for first-aid and medical material. An UNDRO team of five, including three members of the Swiss Disaster Relief Unit, flew to Aden to assess relief needs and co-ordinate assistance. Following consultations with agencies concerned, UNDRO organized a special flight for the transport of relief supplies urgently needed to replenish stocks severely depleted by the civil strife.

46. Total contributions reported to UNDRO were \$US 1,502,410, of which \$US 44,241 were channelled through UNDRO.

Notes

a/ Office of the United Nations Disaster Relief Co-ordinator, "Final report on execution, procedures and experience gained", December 1986.

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**Disaster contribution summary for 1986 of the Office of the
United Nations Disaster Relief Co-ordinator**

Country	Disaster	Month opened	Month closed	Estimated number of persons affected	UNDR emergency grant	United Nations system	Inter- governmental organizations	Governments	League of Red Cross and Societies	Non- governmental organizations	Other	Total	Cash grants through UNDR /
Disasters occurred before 1986 (contribution made in 1986)													
Bangladesh	Cyclone	5.85	10.85	100 000				233 715				233 715	233 716
Ethiopia	Drought/civil strife	4.84	12.86	6 500 000				376 449 124				551 645 098	1 167 683
Viet Nam	Typhoon/floods	10.85	3.86	2 800 000				31 507				31 507	31 507
Disasters occurred in 1986													
Angola	Civil strife/airlift	8.86	8.86	280 000				772 939		73 349		816 288	850 325
Argentina	Floods	4.86	6.86	100 000				80 014		5 000		145 666	
Bolivia	Floods	1.86	5.86	250 000	30 000	135 000	451 782	1 520 524	571 915	431 324	14 000	3 154 545	130 867
Burma	Fire	3.86	4.86	22 087		50 000	98 039	78 411	5 970			232 420	4 000
Cameroon	Volcanic toxic gas	8.86	10.86	10 000		47 898	102 204	2 953 927	386 298	257 497	2 271 642	6 019 466	14 098
Chad	Drought/civil strife	1.86	12.86	1 500 000		750 000	5 680 619	3 218 743	802 381	2 643 740		13 095 483	
Colombia	Volcanic eruption	1.86	8.86	28 000				389 719				389 719	389 719
Colombia	Floods	6.86	10.86	28 000				153 513	98 446	249 180		501 139	
Democratic Yemen	Civil strife	1.86	4.86	500 000	2 000		394 391	803 016	321 312			1 520 719	44 241
El Salvador	Earthquake	10.86	11.86	10 000	30 000	360 000	1 020 267	66 098 874	1 538 950	3 815 186	294 000	73 157 277	125 000
Fiji	Cyclone	4.86	4.86	1 964		9 346	383 178	424 027		1 224		817 775	
Greece	Earthquake	9.86	10.86	25 000		10 000		880 247	184 211			1 074 458	
Haiti	Sequel to civil unrest	4.86	6.86				490 196	14 612 448		90 084		15 192 728	
Haiti	Fire	5.86		2 500				150 553				150 553	
Haiti	Floods in the south	6.86	6.86	86 000	30 000			1 467 694	218 229	137 906		1 853 829	298 266
Jamaica	Floods	6.86	6.86	40 000		669 320	376 294	1 677 156	2 999	39 142		2 764 911	
Lebanon	Armed conflict	10.86			50 000	395 000	6 159 992	13 104 888	82 472			19 792 352	882 382
Madagascar	Cyclone	3.86	4.86	120 000		125 000	717 153	597 455	76 773	218 056		1 734 437	
Mali	Drought	1.86	12.86	1 200 000		2 175 000	9 926 290	12 704 647	326 786	1 498 888		26 631 611	440 000
Mali	Displaced persons	5.86	12.86	7 100		26 000		91 083				117 083	51 055
Mozambique	Civil strife	6.86	12.86	1 848 000			694 881	24 507 244	72 100	9 748 652		35 022 877	1 061 784
Nigeria	Yellow fever	11.86		1 400 000		370 000	958 466	620 588	15 000			1 964 054	
Peru	Floods	3.86	9.86	200 000	25 000	215 174	375 235	1 412 510	173 312	954 517		3 155 748	159 419
Peru	Earthquake	4.86	5.86	600				79 859		80 900		160 759	
Saint Vincent and the Grenadines	Tropical storm	9.86	12.86					200 910				200 910	200 911
Solomon Islands	Cyclone	5.86	7.86	90 000		649 000	505 540	2 100 979	108 376	221 750	118 314	3 703 959	
Sri Lanka	Dam burst	4.86	5.86			15 000		348 189		18 055		381 244	10 695
Sudan	Drought/civil strife	1.86	12.86	5 000 000		20 140 833	47 706 264	43 415 836		5 560 734	3 000 000	119 823 667	100 000
Viet Nam	Typhoon "Wayne"	9.86	10.86	1 000 000		960 000	718 096	753 384		87 332	39 606	2 558 418	
Grand total				23 121 251	167 000	48 326 283	178 484 487	571 941 723	5 385 950	64 251 567	19 517 405	888 074 415	6 190 868

Notes:

Country: disaster-affected countries in which UNDR was involved.

Month opened: month in which a disaster occurred (the contribution recording begun).

Estimated figures: those known at the "month opened" time.

Contributions: dollar-equivalent of cash and in-kind contributions given for disaster relief and reported to UNDR, when possible, estimates used for contributions with no values.

Month closed: month in which last SITREP issued.

a/ These figures represent actual funds received at UNDR during 1986. They could differ from those included in the "Total", as the latter was calculated based on pledges using the exchange rate prevailing at the time the pledges were communicated to UNDR.

ANNEX VI (continued)

Country	Disaster	Month opened	Month closed	Estimated number of persons affected	UNDRO emergency grant	United Nations system	Inter-governmental organizations	Governments	League of Red Cross and Red Crescent Societies	Non-governmental organizations	Other	Total	Cash grants through UNDRO a/
Disasters occurred before 1987 (contribution made in 1987)													
Algeria	Displaced persons	12.86				379 955		246 264				626 219	138 726
Ethiopia	Drought/civil strife	4.84		6 500 000		500 000	3 730 000	20 872 764		580 000		25 682 764 b/	425 009
	Drought/civil strife (airlift)	11.87						3 676 271				3 676 271	3 676 272
Haiti	Floods in the south	6.86	9.86					396 656				396 656	396 656
Lebanon	Armed conflict	10.86	11.87			5 501 420	4 319 738	20 238 365		96 500		30 156 023	443 983
	Appeal by the Secretary-General	12.87		1 400 000				242 424				242 424	242 424
Sudan	Drought/civil strife	8.86	12.87			20 777 928	665 000	2 681 100		733 592		24 857 620	35 000
Disasters occurred in 1987													
Bangladesh	Floods	6.87	11.87	20 900 000	25 000	5 396 920	5 026 809	54 938 898	1 926 294	4 106 198	3 268	71 423 387	346 153
Chad	Displaced persons	2.87	5.87	10 000	30 000	92 433	749 280	5 540 014	343 777	472 500		7 228 004	65 083
China	Pire	5.87	9.87	56 000	30 000	2 167 700	576 369	2 872 544	228 529	5 000		5 850 142	36 746
Comoros	Cyclone	1.87	4.87	35 000				308 550				308 550	
Ecuador	Earthquake	3.87	6.87	26 000	25 000	1 470 000	567 344	4 190 137	620 446	569 077	205 243	7 647 247	319 492
Fiji	Cyclone Raja	1.87	4.87		20 000							20 000	
Guatemala	Floods	10.87	11.87	3 500	10 000		3 500	522 935		96 909		629 844	44 472
Haiti	Torrential rains	7.87	10.87	5 000				141 841				141 841	141 841
Lesotho	Snowstorms	10.87	11.87	100 000	10 000	50 000	49 261	1 188 517	207 647	49 837		1 555 262	
Maldives	Tidal waves	4.87	8.87		10 000	20 000		544 357		10 000		584 357	12 324
Mozambique	Civil strife	1.87		3 529 200		10 232 281	34 205 836	268 620 788	2 398 217	22 064 228	126 599	337 647 949	1 391 588
Philippines	Typhoon Sisang	11.87	12.88	100 000		399 550	620 117	1 877 282	182 019	166 314		3 245 282	
Somalia	Drought	5.87	9.87	5 100 000	30 000	1 357 475		772 778		32 231		2 192 484	
Vanuatu	Cyclone	2.87	5.87	23 000	10 000	403 213	529 981	2 310 517	52 507	23 038		3 329 256	
Viet Nam	Typhoon "Maury"	11.87		360 000	20 000	50 000	2 493 585	48 780		25 000		2 637 365	
Total				38 147 700	190 000	48 798 875	53 533 320	392 231 782	5 959 436	29 030 424	335 110	530 078 947	7 705 769

Notes:

Country: disaster-affected countries in which UNDRO issued situation report(s) or more than three information reports.

Month opened: month in which a disaster occurred.

Month closed: one month after the issuing of last situation or information report.

Estimated figures: those known at the "month opened" time.

Contributions: dollar-equivalent of cash/in-kind contributions given for disaster relief and reported to UNDRO; when possible, estimates used for contributions with no values.

a/ These figures represent actual funds received at UNDRO during 1987. They could differ from those included in "Total" as the latter was calculated based on pledges using the exchange rate prevailing at the time the pledges were communicated to UNDRO.

b/ This represents only non-food portion. Value of food aid contributions not yet received.

ANNEX VII

Total contributions by donor country (all disasters) for
which value was reported to the Office of the United
Nations Disaster Relief Co-ordinator (1986-1987)

(in United States dollars)

Donor country	1986	1987	Total 1986-1987
Argentina	180 000	110 000	290 000
Australia	10 620 896	16 224 521	26 845 417
Austria	5 570 810	1 801 512	7 372 322
Bangladesh	-	618 638	618 638
Bahrain	100 000	-	100 000
Barbados	5 000	-	5 000
Belgium	602 195	1 863 395	2 465 590
Bhutan	-	50 000	50 000
Bolivia	-	3 800	3 800
Botswana	-	1 176 000	1 176 000
Brazil	(++)	2 197	2 197
Canada	29 794 390	18 295 230	48 089 620
Chile	55 000	600 000	655 000
China	4 847 256	86 954	4 934 210
Colombia	-	(++)	(++)
Cuba	-	(++)	(++)
Cyprus	-	1 000	1 000
Czechoslovakia	-	697 179	697 179
Democratic Kampuchea	667	-	667
Denmark	1 782 066	11 541 136	13 323 202
Dominican Republic	1 499	1 000	2 499
Fiji	9 346	(++)	9 346
Finland	1 242 206	3 406 292	4 648 498
France	7 778 368	4 473 297	12 251 665
Gabon	287 356	-	287 356
German Democratic Republic	-	6 194 302	6 194 302
Germany (Federal Republic of)	14 624 197	11 617 458	26 241 655
Greece	182 500	-	182 500
Guatemala	30 000	-	30 000
Holy See	65 000	10 000	75 000
Hong Kong	-	640 205	640 205
Hungary	-	11 419	11 419
India	1 544 200	1 415 194	2 959 394
Indonesia	10 000	-	10 000

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Donor country	1986	1987	Total 1986-1987
Iran (Islamic Republic of)	386 960	-	386 960
Iraq	-	(++)	(++)
Ireland	1 067 114	240 410	1 307 524
Israel	(++)	18 890	18 890
Italy	39 001 671	50 366 626	89 368 297
Jamaica	-	1 000	1 000
Japan	7 548 856	16 259 639	13 808 495
Kenya	-	396 000	396 000
Kuwait	34 602	(++)	34 602
Lao People's Democratic Republic	8 658	-	8 658
Luxembourg	192 858	172 840	365 698
Mauritius	(++)	-	(++)
Mexico	(++)	-	(++)
Morocco	-	(++)	(++)
Nauru	14 815	6 623	21 438
Nepal	-	90 000	90 000
Netherlands	5 679 035	28 646 384	34 325 419
New Zealand	346 711	363 393	710 104
Nigeria	765 000	(++)	765 000
Norway	5 607 137	11 744 844	17 351 981
Pakistan	-	579 374	579 374
Papua New Guinea	155 208	-	155 208
Peru	(++)	-	(++)
Philippines	-	(++)	(++)
Poland	2 590 000	(++)	2 590 000
Portugal	-	3 000 000	3 000 000
Republic of Korea	29 055	180 000	209 055
Saudi Arabia	-	10 000 000	10 000 000
Singapore	-	(++)	(++)
Spain	126 018	283 403	409 421
Sri Lanka	1 700	(++)	1 700
Sweden	24 952 459	50 092 049	75 044 508
Switzerland	6 874 422	1 704 966	8 579 388
Taiwan	200 000	50 000	250 000
Thailand	10 000	-	10 000
United Kingdom of Great Britain and Northern Ireland	18 011 997	23 444 473	41 456 470
United States of America	241 302 448	113 061 628	354 364 076
Union of Soviet Socialist Republics	133 000 000	361 711	133 361 711

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Donor country	1986	1987	Total 1986-1987
Venezuela	(++)	225 000	225 000
Yemen Arab Republic	(++)	-	(++)
Yugoslavia	-	101 800	101 800
Zaire	100 000	-	100 000
Zimbabwe	4 602 047	-	4 602 047
 Total, Governments	 571 941 723	 392 231 782	 964 173 505
 European Economic Community	 178 484 487	 53 533 320	 232 017 807
 GRAND TOTAL	 750 426 210	 445 765 102	 1 196 191 312

Notes:

(++) Contribution(s) reported to UNDRO without corresponding value. Many other countries reported contributions with no corresponding value; these, of course, could not be included in the totals for those countries listed above.